Lab 03: DHCP and NAT Configuration

Duration: 2 hours 30 minutes | In-Lab Assessment
Tools: Cisco Packet Tracer

Objectives

By the end of this lab, students will be able to:

- Set up DHCP both via a dedicated server and via a Cisco router.
- Inspect DHCP bindings and troubleshoot IP lease assignments.
- Configure Static, Dynamic, and Port-Address Translation (PAT) NAT on Cisco routers.
- Verify NAT translations and statistics.
- Understand and configure Access Control Lists (ACL).

Part 1: DHCP Theory and Configuration

Part A: Dynamic Host Configuration Protocol (DHCP)

DHCP is a network management protocol that automates the assignment of IP addresses and other parameters (such as Subnet Mask, Default Gateway, and DNS), allowing devices to communicate on a network without manual configuration.

- **Operation:** The DHCP server selects an IP from a predefined pool and assigns it to a client upon request.
- **Lease:** Addresses are granted for a limited time (a lease). Clients must renew this lease periodically to retain connectivity.
- **Relay Agents:** Since DHCP relies on broadcast messages, which do not cross routers, a *DHCP Relay Agent* is required if the Server and Client are on different nets/subnets. The agent forwards client requests to the remote server.

Part B: Configure DHCP (using server)

A router of device model 2811, two switches of device model 2960, two PCs, two laptops, and one server have been used in the sample topology shown in Figure 4 for the DHCP configuration.

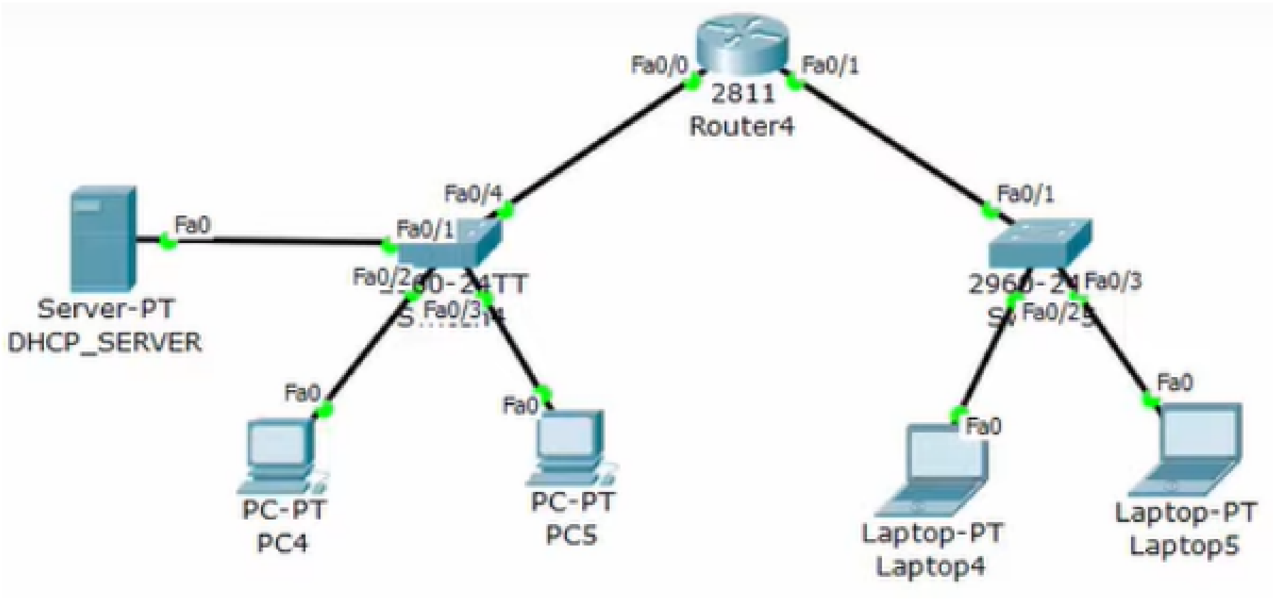

Figure 1: A sample network topology for the configuration of DHCP.

1. **Configure DHCP Server**

```
IP Address: 192.168.1.2
Default: 192.168.1.1
```

2. **Make DHCP Pools**

Go to Services and then DHCP, and change the following fields.

```
Pool Name: dotONEnetwork
Default: 192.168.1.1
Start IP: 192.168.1.3
Max Number: 20
```

Select **Add** to add the Address Pool to the server. Select **Save** to save the modification to the server.
To add another pool to the server, change the information as follows and then hit **Add** and **Save** buttons again.

```
Pool Name: dotTWOnetwork
Default: 192.168.2.1
Start IP: 192.168.2.2
Max Number: 20
```

Do not forget to turn on the DHCP server.

3. **Configure R1 Interfaces**

```
R1(config)#int fa0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# ip helper-address 192.168.1.2
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)# ip address 192.168.2.1 255.255.255.0
```

```
R1(config-if)# ip helper-address 192.168.1.2
R1(config-if)#no shutdown
R1(config-if)# end
R1# copy running-config startup-config
```

4. **Configure all the PCs**
   Just click DHCP and the server will do the rest.

5. **Verify**

   ```
   Ping PC1 from PC0
   ```

Part C: Configure DHCP (using router)

The exact network topology with the same device models, as shown in Figure 4, has been used for this configuration.

Configure the DHCP server in the router instead of a server.

```
R1(config)# ip dhcp pool dotONEnetwork
R1(dhcp-config)# default-router 192.168.1.1
R1(dhcp-config)# network 192.168.1.0 255.255.255.0
R1(dhcp-config)# exit

R1(config)# ip dhcp pool dotTWOnetwork
R1(dhcp-config)# default-router 192.168.2.1
R1(dhcp-config)# network 192.168.2.0 255.255.255.0
R1(dhcp-config)# exit
```

The rest of the configuration, i.e., Configure R1 Interfaces, Configure all the PCs, and Verify are same as the section .

Use **show ip dhcp binding** inside the router to see the status of the configured DHCP.

Part 2: NAT and ACL Theory and Configuration

Part A: Standard Access Control Lists (ACL)

An ACL is a sequential list of "permit" or "deny" statements applied to an interface. In this lab, we focus on **Standard IPv4 ACLs** (numbered 1–99), which filter traffic based solely on the **Source IP**.

Every ACL has an invisible "Deny All" rule at the very bottom. If a packet does not match any of your specific rules, it is **discarded**.
Solution: If you want to filter specific traffic but allow everything else, you must explicitly add `permit any` at the end of your list.

Configuration Steps

Step 1: Define the Rules

Use a number between 1 and 99. You must specify the source IP and a wildcard mask (inverse subnet mask).

```
! Format: access-list <1-99> {permit|deny} <source_ip> <wildcard>
Router(config)# access-list 10 deny 192.168.1.5 0.0.0.255
Router(config)# access-list 10 permit any
```

Step 2: Apply to an Interface

An ACL has no effect until applied to an interface's inbound or outbound direction.

```
! Format: ip access-group <number> {in|out}
Router(config)# interface g0/0
Router(config-if)# ip access-group 10 out
```

Step 3: Verify

```
Router# show access-lists
```

Part B: NAT Overview

NAT allows private IP networks to connect to the internet by translating private addresses to global public addresses.

1. **Static NAT:** One-to-One mapping (1 Private IP $\leftrightarrow$ 1 Public IP). Used for servers.
2. **Dynamic NAT:** Many-to-Many mapping. Maps a private IP to the first available public IP from a pool.
3. **PAT (Overload):** Many-to-One mapping. Maps multiple private IPs to a single public IP using different source port numbers.

Part C: Prerequisites: Defining Interfaces

For any NAT configuration, you must define the boundaries of the network.

```
Router(config)# interface fa0/0 ! Interface facing LAN
Router(config-if)# ip nat inside
Router(config)# interface fa0/1 ! Interface facing Internet/ISP
Router(config-if)# ip nat outside
```

Part D: Configuration A: Static NAT

Goal: Map local server `10.0.0.5` permanently to public IP `50.0.0.5`.

```
Router(config)# ip nat inside source static 10.0.0.5 50.0.0.5
```

Part E: Configuration B: Dynamic NAT

Goal: Allow internal subnet `192.168.10.0/24` to share a pool of public IPs (`50.0.0.1` to `50.0.0.3`).

```
! Step 1: Define traffic to be translated using an ACL
Router(config)# access-list 1 permit 192.168.10.0 0.0.0.255

! Step 2: Define the pool of Public IPs
Router(config)# ip nat pool MY_POOL 50.0.0.1 50.0.0.3 netmask 255.255.255.252

! Step 3: Link the ACL to the Pool
Router(config)# ip nat inside source list 1 pool MY_POOL
```

Part F: Configuration C: PAT (NAT Overload)

Goal: Allow hundreds of users to access the internet using a single Public IP (on the outgoing interface).

```
! Step 1: Define traffic (ACL)
Router(config)# access-list 1 permit 192.168.10.0 0.0.0.255

! Step 2: Enable Overload on the outside interface
Router(config)# ip nat inside source list 1 interface fa0/1 overload
```

Part G: Configure ACL

A router of device model 2911, three switches of device model 2960, and three PCs have been used in the sample topology shown in Figure 5 for the ACL configuration.

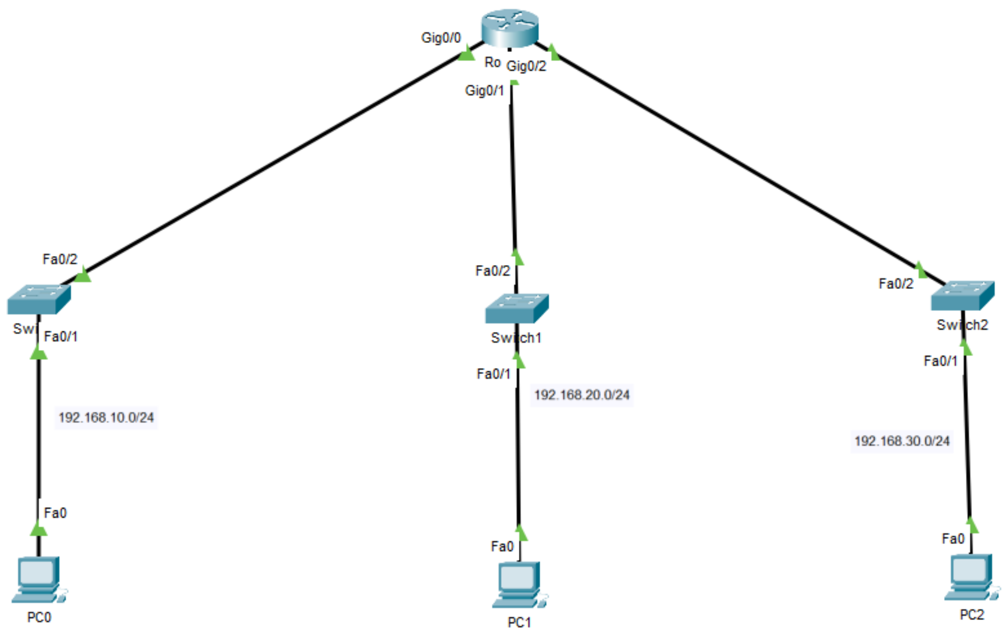

Figure 2: A sample network topology for the configuration of ACL.

1. **Define and Apply ACL**

```
Router(config)# access-list 1 deny 192.168.10.0 0.0.0.255
Router(config)# access-list 1 permit any
Router(config)# interface g0/2
Router(config-if)# ip access-group 1 out
```

Part H: Configure Static NAT

Two routers of device model 2811, one switch of device model 2960, two PCs, and one server have been used in the sample topology shown in Figure 6.

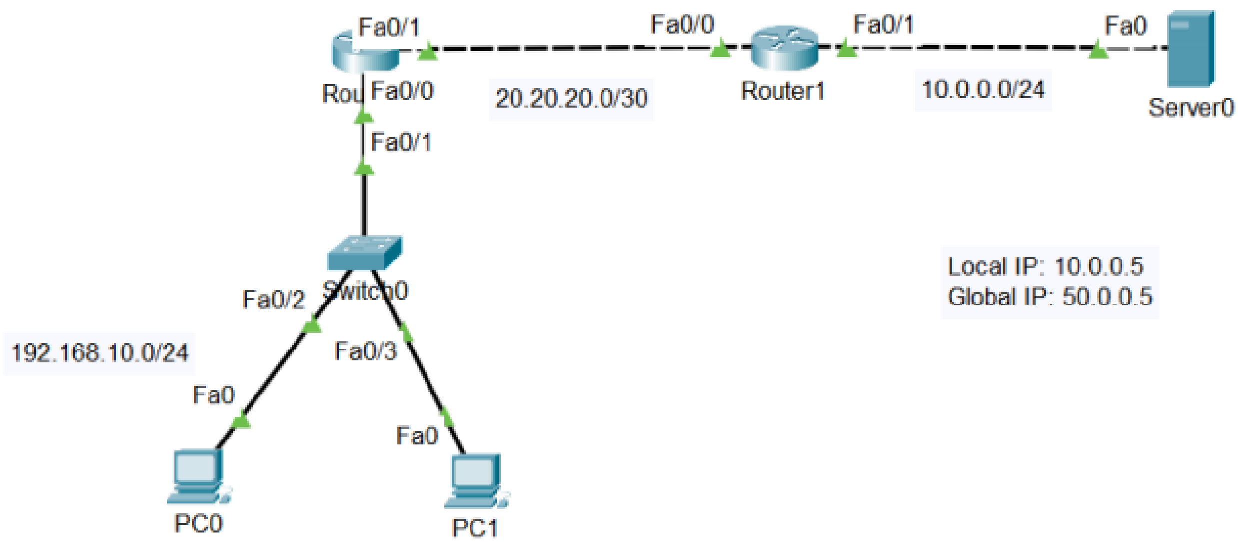

Figure 3: A sample network topology for the configuration of NAT.

1. **Enable static NAT inside Router1**

```
Router(config)# ip nat inside source static 10.0.0.5 50.0.0.5
Router(config)# int fa0/1
Router(config-if)# ip nat inside
Router(config)# int fa0/0
Router(config-if)# ip nat outside
```

2. **Verify NAT**

```
Router# show ip nat translations
```

Part I: Configure Dynamic NAT

Using the topology in Figure 6:

1. **Enable dynamic NAT inside Router1**

```
Router(config)# access-list 1 permit 192.168.10.0 0.0.0.255
Router(config)# ip nat pool NAT_Pool 50.0.0.1 50.0.0.3 netmask 255.255.255.252
Router(config)#ip nat inside source list 1 pool NAT_Pool
Router(config)# int fa0/1
Router(config-if)# ip nat inside
Router(config)# int fa0/0
Router(config-if)# ip nat outside
```